Chapter 7 — Functions

A **function** is a named, reusable block of code that takes inputs, does one job, and (usually) returns a result. Beyond ten-line scripts, everything is built from functions, and many "advanced" features — decorators (functions wrapping functions to add behaviour, 7.6), callbacks (functions handed to other code to call later), `sorted(key=...)`, caching (remembering results) — are just functions passed around as values.

This chapter goes from `def` to closures (functions remembering variables from where they were created), decorators and recursion (a function calling itself). Scripts are in `code/`, run with Python 3.14 (3.12-compatible); output is real.

What you'll learn

#	Topic	Script
7.1	Defining and calling, `return`, `None`, `pass`	`01_defining_and_arguments.py`
7.2	Every kind of argument: positional, keyword, default, `*args`, `**kwargs`, `/`, `*`, unpacking, the mutable-default trap, multiple return values	`01_defining_and_arguments.py`
7.3	Scope and namespaces: LEGB, `global`, `nonlocal`	`02_scope_and_closures.py`
7.4	Closures and the late-binding trap	`02_scope_and_closures.py`
7.5	First-class and higher-order functions, `lambda`, `map` / `filter`, `key=`	`03_first_class_and_functools.py`
7.6	Decorators and `functools`: `wraps`, `partial`, `reduce`, `lru_cache`, `cache`	`03_first_class_and_functools.py`
7.7	Recursion, the call stack, the recursion limit, memoization	`04_recursion.py`
7.8	Docstrings and type annotations	`05_docstrings_annotations.py`

7.1 Defining and calling functions

What it is

Define with `def`, a name, parentheses holding the **parameters**, and a colon; the indented lines are the **body**. Defining doesn't run it — it creates a function object bound to the name. **Call** it with its name plus parentheses holding **arguments**.

> **Parameter vs argument.** The *parameter* is the name in the definition (`def cube(x)` → `x`); the *argument* is the value at the call (`cube(3)` → `3`). Interviewers notice mix-ups.

Why it matters

1. **Modularity** — small pieces with one job, written and debugged alone.
2. **Reuse** — write once, call anywhere (or from other programs via modules, Chapter 8); fix a bug once.
3. **Scope** — names inside are private, so parts don't trample each other's variables (7.3).
4. **Readability and teamwork** — `send_invoice(order)` documents intent; people can own different functions.

How it works

1. `def` compiles the body to bytecode, creates a *function object*, evaluates default values (important in 7.2), binds the name.
2. A call creates a fresh **frame** (workspace of local variables), binds parameters to arguments, runs the body.
3. `return value` ends the call and hands `value` back; falling off the end or a bare `return` gives `None`.

4. The frame is discarded (unless a closure keeps part alive, 7.4).

`pass` does nothing — the placeholder where a block is required (empty function, class, `if`). `...` (Ellipsis) is used the same way in stubs.

```
def say_hi():
    """A function with no parameters and no return statement."""
    print("hi from say_hi()")

def cube(x):
    """Return x cubed."""
    return x ** 3

def todo():
    pass  # placeholder body - keeps the syntax valid

say_hi()
print("say_hi() returned:", say_hi())
print("cube(3) =", cube(3))
print("todo() returned:", todo())
```

Output:

```
hi from say_hi()
hi from say_hi()
say_hi() returned: None
cube(3) = 27
todo() returned: None
```

`say_hi()` printed twice (plain call, then inside `print`), and returned `None` — a function that only *prints* still *returns* `None`.

Trade-offs: print vs return

	`print(...)` inside	`return ...`
Result usable by caller	No — goes to the screen	Yes — stored, tested, combined
Testable	Only by capturing stdout (the console stream)	Directly (`assert cube(3) == 27` — `assert` raises if false)
Good for	Debugging, CLI output at the edges	Almost all real logic

Common mistakes

- **Missing parentheses**: `result = cube` stores the function; `cube(3)` calls it.
- **Printing instead of returning** → `x = f()` is `None`.
- **Code after** `return` in the same block never runs.
- **Calling before defining** at module level — the `def` must have executed first (inside other functions it's fine; lookup happens at call time).

Real-world examples

Libraries are collections of functions (`len`, `open`, `json.dumps`, `requests.get`, `numpy.mean`); Flask and FastAPI map each URL to a plain function ("view"/"endpoint") returning the response.

In the interview

30-second answer: "A function is defined with `def`; defining creates a function object bound to a name, and calling it creates a new frame with its own local variables. `return` sends a value back; with no `return` the result is `None`. Functions give modularity, reuse and a private scope."

- *Are functions objects?* → Yes; they have attributes (`__name__`, `__doc__`, `__defaults__`) and can be passed around (7.5).
- `pass`*?* → No-op placeholder where syntax needs a statement.

Self-check

1. `x = print("a")` stores? → `None`.

TL;DR

`def name(params):` + indented body. Defining ≠ running. `return` gives a value back, otherwise `None`. `pass` fills an empty body.

7.2 Function arguments — every kind

What it is

At the **call site** (where the function is called) pass arguments **positionally** (by order) or **by keyword** (by name). In the **definition** add **defaults**, collect extras with `*args`/`**kwargs`, and force a style with `/` (positional-only) and `*` (keyword-only).

The full parameter list, in the only allowed order:

```
def f(pos_only, /, pos_or_kw, *args, kw_only, **kwargs): ...
#     ^^^^^^^^     ^^^^^^^^^^  ^^^^^  ^^^^^^^  ^^^^^^^^
#     before /     normal      extra  after *  extra keywords
#                              positionals     (dict)
#                              (tuple)
```

Kind	Declared as	Called as	Notes
Positional-or-keyword	`def f(a, b)`	`f(1, 2)` or `f(b=2, a=1)`	The default kind
Default	`def f(a, b=10)`	`f(1)` or `f(1, 5)`	Defaulted params come after non-defaulted (positional part)
Variable positional	`def f(*args)`	`f(1, 2, 3)`	`args` is a **tuple**
Variable keyword	`def f(**kwargs)`	`f(x=1, y=2)`	`kwargs` is a **dict** (insertion-ordered)
Positional-only	`def f(a, b, /)`	`f(1, 2)` only	Python 3.8+ (PEP 570)
Keyword-only	`def f(*, a, b)` or after `*args`	`f(a=1, b=2)` only	Can be required (no default)

Why it matters

Good argument design makes calls **self-documenting and hard to misuse**: `connect("db", 5432, True, False)` is a riddle; `connect("db", port=5432, ssl=True, retry=False)` explains itself. Defaults keep simple calls simple, `*args`/`**kwargs` enable flexible wrappers (functions calling another and adding something around it), and `/`/`*` let library authors rename parameters later.

How it works — binding arguments to parameters

1. Positional arguments fill parameters left to right; extras go to `*args` (or `TypeError`).
2. Keyword arguments match by name; unknown names go to `**kwargs` (or `TypeError`).
3. Filled twice (position *and* keyword) → `TypeError: got multiple values`.
4. Empty parameters take their **default**; none → `TypeError: missing ... required argument`.

At the call, positionals precede keywords: `f(1, b=2)` fine, `f(a=1, 2)` → `SyntaxError`.

Positional and keyword arguments

Positional arguments match purely by order — swap two and you silently get wrong results. **Keyword** arguments name the parameter, so order stops mattering and calls read clearly — best with many or optional parameters.

Default arguments

A **default** makes a parameter optional; a supplied argument overrides it.

```
def book_flight(origin, destination, seat="economy", *, meal="standard"):
    """origin/destination: positional-or-keyword; seat: default; meal: keyword-only."""
    return f"{origin}->{destination} [{seat}, meal={meal}]"

print(book_flight("SFO", "JFK"))                                  # defaults used
print(book_flight(destination="JFK", origin="SFO", seat="aisle"))  # any order by name
print(book_flight("SFO", "JFK", meal="vegan"))                    # keyword-only
book_flight("SFO")                          # missing arg
book_flight("SFO", "JFK", "aisle", "vegan")  # meal given positionally
book_flight("SFO", origin="LAX")             # same arg twice
```

Output (errors caught by a helper in the script):

```
SFO->JFK [economy, meal=standard]
SFO->JFK [aisle, meal=standard]
SFO->JFK [economy, meal=vegan]
  missing arg: TypeError: book_flight() missing 1 required positional argument: 'destination'
  meal given positionally: TypeError: book_flight() takes from 2 to 3 positional arguments but 4 were given
  same arg twice: TypeError: book_flight() got multiple values for argument 'origin'
```

Positional-only (`/`) and keyword-only (`*`)

- **Before `/`**: by position only. Names become internal, so they can be renamed later, and `value` can't collide with a `**kwargs` key. Built-ins do this: `len(obj, /)`, `abs(x, /)`, `divmod(x, y, /)` — `len(obj=[1])` is a `TypeError`.
- **After `*`** (or `*args`): by name only. Use for flags and non-obvious options: `sorted(iterable, /, *, key=None, reverse=False)` forbids `sorted(xs, None, True)`. Keyword-only parameters may be required.

```
def clamp(value, low, high, /):
    return max(low, min(value, high))

print("clamp(15, 0, 10) =", clamp(15, 0, 10))
clamp(value=15, low=0, high=10)
```

```
clamp(15, 0, 10) = 10
  clamp(value=15, ...): TypeError: clamp() got some positional-only arguments passed as keyword arguments: 'value, low, high'
```

Variable-length arguments: `*args` and `**kwargs`

`*args` gathers **extra positionals** into a tuple; `**kwargs` gathers **extra keywords** into a dict. The names are convention — the stars do the work.

```
def total(*numbers, start=0):
    print("  numbers is a", type(numbers).__name__, numbers)
    return sum(numbers, start)

def make_tag(tag, /, text, **attrs):
    attr_str = "".join(f' {k}="{v}"' for k, v in attrs.items())
    return f"<{tag}{attr_str}>{text}</{tag}>"

print("total(1, 2, 3) =", total(1, 2, 3))
print("total() =", total())
print(make_tag("a", "docs", href="https://docs.python.org", target="_blank"))
```

```
  numbers is a tuple (1, 2, 3)
total(1, 2, 3) = 6
  numbers is a tuple ()
total() = 0
<a href="https://docs.python.org" target="_blank">docs</a>
```

E.g. `configure(host="db1", port=5432, timeout="30s")` receives `{"host": "db1", "port": 5432, "timeout": "30s"}` and loops over `.items()`.

Unpacking at the call site

In reverse: `*iterable` spreads positionals, `**mapping` spreads keywords.

```
nums = [4, 5, 6]
opts = {"seat": "window", "meal": "kosher"}
print("total(*nums) =", total(*nums))
print(book_flight("SFO", "JFK", **opts))
```

```
  numbers is a tuple (4, 5, 6)
total(*nums) = 15
SFO->JFK [window, meal=kosher]
```

Returning several values

A function returns **one** object; `return a, b, c` builds a tuple that the caller unpacks.

```
def min_max_mean(values):
    return min(values), max(values), sum(values) / len(values)

result = min_max_mean([3, 9, 6])
print("raw result:", result, type(result).__name__)
lo, hi, avg = result
```

```
raw result: (3, 9, 6.0) tuple
lo=3 hi=9 avg=6.0
```

Beyond ~3 values, return a `dataclass` or `typing.NamedTuple` (records with named fields, Chapters 5 and 10) so callers write `stats.mean`.

The mutable-default pitfall

Defaults are evaluated once — when `def` runs — not per call. A mutable default (list, dict, set) is *one shared object*, so changes pile up.

```
def append_bad(item, bucket=[]):          # BUG: default evaluated once
    bucket.append(item)
    return bucket

def append_good(item, bucket=None):       # FIX: None sentinel
    if bucket is None:
        bucket = []
    bucket.append(item)
    return bucket

print(append_bad("a"), append_bad("b"))
print("default is now:", append_bad.__defaults__)
print(append_good("a"), append_good("b"))
```

```
['a', 'b'] ['a', 'b']
default is now: (['a', 'b'],)
['a'] ['b']
```

Both `append_bad` calls returned the *same* list; `__defaults__` shows the mutated default. Fix with the `None` **sentinel** (placeholder meaning "nothing passed"): create the object inside. The same rule freezes

`def log(msg, when=datetime.now())` at definition time. `dataclasses` refuses bare mutable defaults and asks for `field(default_factory=list)`.

Trade-offs

Choice	Benefit	Cost
Positional	Short calls	Order-dependent, easy to swap by mistake
Keyword	Readable, order-free	Longer; names become public API others rely on
Keyword-only (`*`)	Forces clarity for flags/options	Slightly more typing
Positional-only (`/`)	Free to rename; no clash with `**kwargs`	Callers can't use names for clarity
`*args` / `**kwargs`	Flexible wrappers, forwarding	Hides the real signature (parameter list) from readers, IDEs and type checkers

When not to use it

- Don't use `**kwargs` when you know the options — explicit keywords error on typos (`timout=5`) and get IDE help; `**kwargs` swallows the typo.
- Don't stack boolean positional flags (`f(x, True, False)`) — make them keyword-only.

Common mistakes

- Mutable default (`=[]`, `={}`) — use `None`.
- `def f(a=1, b):` → `SyntaxError`; `f(a=1, 2)` → `SyntaxError`.
- `*args` is a **tuple** — `list(args)` to modify.
- `return a, b` is one tuple, not "two return values".

Real-world examples

- `print(*objects, sep=" ", end="\n", file=None, flush=False)` — `*objects` plus keyword-only options.
- `requests.get(url, params=None, **kwargs)` forwards `timeout=`, `headers=`.
- Matplotlib plotting calls pass `**kwargs` (`color=`, `linewidth=`) to underlying line/patch objects.

In the interview

> **30-second answer:** "Arguments bind positionally by order or by keyword by name. Defaults make parameters optional but are evaluated once at definition time, so never use a mutable default — use `None`. `*args` collects extra positionals into a tuple, `**kwargs` extra keywords into a dict. Parameters before `/` are positional-only, after `*` keyword-only."

- *Why is `def f(x=[])` dangerous?* → One list, created once, shared by every defaulted call.
- *Why positional-only in a library?* → Names stay private (renameable) and a `**kwargs` key may share the name.
- *Pass a list as separate arguments?* → `f(*my_list)`; dict as keywords → `f(**my_dict)`.

Self-check

1. `def f(a, /, b, *, c)` — which work: `f(1, 2, 3)`, `f(1, b=2, c=3)`, `f(a=1, b=2, c=3)`? → Only `f(1, b=2, c=3)` (and `f(1, 2, c=3)`).

TL;DR

Positional = order, keyword = name. Defaults evaluated **once** → `None` for mutables. `*args` → tuple, `**kwargs` → dict; `*`/`**` at the call unpack. `/` positional-only before, `*` keyword-only after. Multiple returns = one tuple.

7.3 Scope and namespaces — LEGB, `global`, `nonlocal`

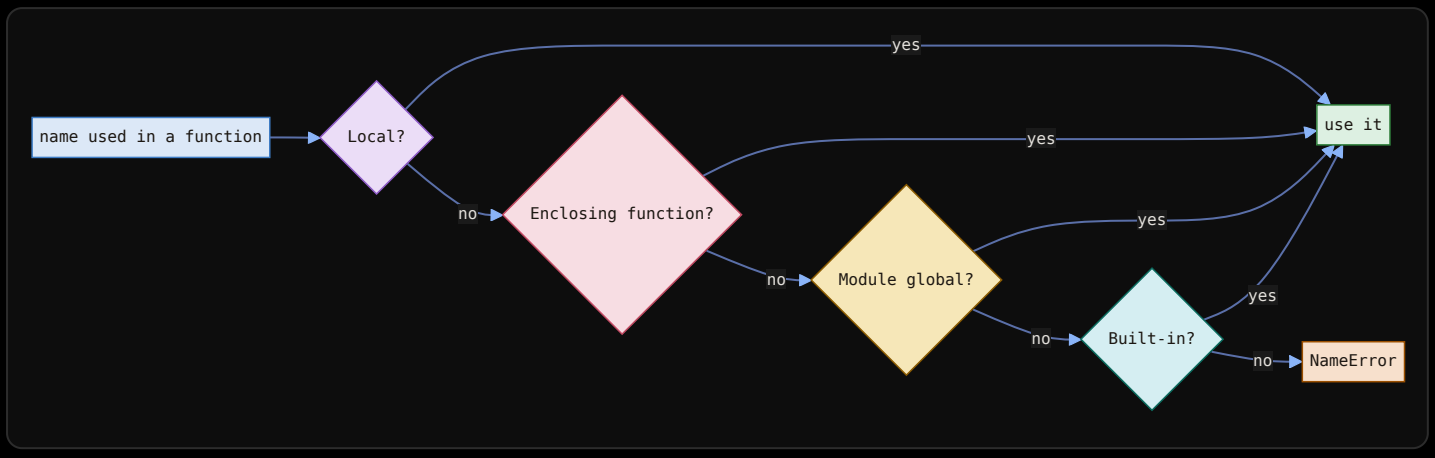
What it is

A **namespace** maps names to objects — usually literally a dict (`globals()` shows the module's). Names are unique within one namespace but can repeat across namespaces. A name's **scope** is the code region where it can be used directly.

Python resolves a bare name with the **LEGB rule**, stopping at the first hit:

Letter	Scope	Where it comes from	Lifetime
L	Local	Names assigned in the current function (incl. parameters)	The call
E	Enclosing	Locals of *outer* functions (nested functions)	While referenced (closures)
G	Global	Top-level names of the **module** (file)	Until program end / unload
B	Built-in	`len`, `print`, `ValueError` … from `builtins`	Whole program

Not found → `NameError`.



Why it matters

Scope makes functions safe to combine — one function's `total` can't clobber another's. Misunderstanding it causes the confusing `UnboundLocalError`.

How it works — the key rule

Locality is decided at compile time for the whole body: if a function *assigns* a name anywhere (`=`, `+=`, `for x in`, `import x`, `def x`), that name is local throughout — unless declared `global` or `nonlocal`. *Reading* alone doesn't make it local, so reading globals needs no declaration.

1. `global name` → assignments rebind the **module-level** variable.
2. `nonlocal name` → assignments rebind the **nearest enclosing function's** variable (must exist there; module globals don't count). Added in Python 3.
3. Only functions (and classes, comprehensions, lambdas) create scopes — **not** `if`, `for`, `while`, `with`; a variable assigned inside an `if` is visible after it.

```python
level = "global"                    # G

def outer():
    level = "enclosing"             # E for inner()
    def inner():
        level = "local"             # L
        print("  inner sees:", level)
    inner()
    print("  outer sees:", level)

def reads_global():
    print("  reads_global sees:", level)   # no assignment here -> global

counter = 0

def broken_increment():
    counter += 1                    # assignment makes counter LOCAL -> error

def global_increment():
    global counter
    counter += 1
```

```
== LEGB lookup ==
  inner sees: local
  outer sees: enclosing
  reads_global sees: global
  B (built-in) len is builtins.len: True

== assigning without global ==
  UnboundLocalError: cannot access local variable 'counter' where it is not associated with a value

== global ==
  counter = 2
```

`counter += 1` is `counter = counter + 1`: the assignment makes `counter` local, so reading it first raises `UnboundLocalError` (a `NameError` subclass). `nonlocal` does the same one level in (`make_counter`, 7.4).

Trade-offs

Shared state via	Good	Bad
Pass in / return values	Explicit, testable, thread-friendly	More parameters
`global`	Quick for scripts	Hidden coupling; any function can change it; hard to test
`nonlocal` / closure	State private to one function family	Less obvious than a class
A class with attributes	Clear home for state + behaviour	More code

When not to use it

Avoid `global` except for true module-level config or caches in small scripts. Mutating a global *object* (`CACHE[key] = v`) needs no `global` — only rebinding the *name* does — but it's still shared mutable state.

Common mistakes

- `UnboundLocalError` from `x += 1` on a global — `global x`, or better, return the new value.
- **Shadowing built-ins**: `list = [1, 2]` then `list('xy')` → `'list' object is not callable` (likewise `dict`, `id`, `input`, `sum`, `type`).
- A class body is not an enclosing scope for its methods — use `self.x` (`self` = the object the method was called on, Chapter 10) or `ClassName.x`.
- Comprehension variables don't leak (own scope); a plain `for` variable does.

Real-world examples

- `logger = logging.getLogger(__name__)` at module level, only *read* in functions.
- `MAX_RETRIES = 3` constants: global, UPPER_CASE (PEP 8), treated as read-only.

In the interview

> **30-second answer:** "Python looks names up via LEGB: local, enclosing, global, built-in. Any assignment in a function makes that name local for the whole function, which is why `x += 1` on a global raises `UnboundLocalError`. `global` rebinds module-level names, `nonlocal` rebinds the nearest enclosing function's name. Blocks like `if` and `for` don't create scopes."

- *`global` needed to append to a global list?* → No — `.append` mutates, not rebinds.
- *`global` vs `nonlocal`?* → Module scope vs nearest enclosing *function* scope (must exist there).
- *Why does `def f(): print(x); x = 1` fail with a global `x`?* → The later assignment makes `x` local for the whole body.
- *Can `nonlocal` target a module variable?* → No — `SyntaxError: no binding for nonlocal ... found`.

TL;DR

Lookup **L → E → G → B**. Assignment anywhere in a function ⇒ local throughout. `global` for module names, `nonlocal` for outer-function names; prefer passing and returning values.

7.4 Closures

What it is

A **closure** is an inner function that *remembers* variables from the enclosing scope after the outer function has returned. Those are **free variables**, stored in "cells" on the function (`fn.__closure__`).

Why it matters

Closures build **functions from functions** — configured functions (`make_multiplier(3)`), counters with private state, callbacks carrying context — without a class. Decorators (7.6) are built on them.

How it works

1. `make_multiplier(3)` runs; `factor = 3` is local.
2. Inner `multiply` reads `factor` → the compiler makes it a free variable in a cell.
3. `multiply` is returned; the frame ends but the cell survives.
4. Each `make_multiplier` call creates a **new** cell — `double` and `triple` are independent.
5. *Rebinding* (not just reading) a captured variable needs `nonlocal`.

```python
def make_counter():
    count = 0
    def step():
        nonlocal count              # rebind the enclosing variable
        count += 1
        return count
    return step

def make_multiplier(factor):
    def multiply(x):
        return x * factor           # 'factor' is captured (a free variable)
    return multiply

c1, c2 = make_counter(), make_counter()
print("  c1:", c1(), c1(), c1(), " c2:", c2())
double, triple = make_multiplier(2), make_multiplier(3)
print("  double(5) =", double(5), " triple(5) =", triple(5))
print("  free vars of double:", double.__code__.co_freevars,
      "cell value:", double.__closure__[0].cell_contents)
```

```
  c1: 1 2 3  c2: 1
  double(5) = 10  triple(5) = 15
  free vars of double: ('factor',) cell value: 2
```

The late-binding trap

Closures capture **variables, not values**; the value is read when the inner function *runs*.

```python
funcs = [lambda: i for i in range(3)]
print([f() for f in funcs])         # [2, 2, 2]  - all see the final i
funcs = [lambda i=i: i for i in range(3)]
print([f() for f in funcs])         # [0, 1, 2]  - default captures the current value
```

`i=i` works *because* defaults are evaluated at definition (7.2); `functools.partial` is cleaner.

Trade-offs

	Closure	Class
Code size	Tiny	Larger
Number of operations	Best for one (a callable)	Many methods
Inspectability	State hidden in cells	Attributes visible, easy to debug

When not to use it

Several operations (increment, reset, read), or state others must inspect/serialise (convert to bytes/text) → write a class. Closures can't be pickled by `pickle` (Python's object-serialisation module).

Common mistakes

- Late binding in loops.
- Missing `nonlocal` → `UnboundLocalError` on `count += 1`.

Real-world examples

- Every decorator (`@functools.lru_cache`, Flask's `@app.route("/")`) returns a closure.
- GUI callbacks (Tkinter `command=`) use closures or `partial` to carry which button was pressed.

In the interview

> **30-second answer:** "A closure is a nested function that keeps references to variables of its enclosing scope after that scope has returned. It captures variables, not values, so lambdas in a

loop all see the last value unless you bind with a default argument or `partial`. `nonlocal` lets it rebind the captured variable."

- *Where are captured variables stored?* → Cells in `fn.__closure__`; names in `fn.__code__.co_freevars`.
- *Fix the loop-lambda bug?* → `lambda i=i: ...` or `functools.partial(f, i)`.

TL;DR

Inner function + remembered outer variables = closure. Captures variables (late binding); `nonlocal` to rebind. Foundation of decorators.

7.5 First-class and higher-order functions, `lambda`, `map`/`filter`, `key=`

What it is

Functions are **first-class objects**: assign, store in lists/dicts, pass, return. A **higher-order function** takes or returns a function (`map`, `sorted`, decorators, `make_multiplier`). A `lambda` is an anonymous single-expression function: `lambda x: x * 2` ≈ `def` returning `x * 2`, but no statements, no annotations, name `<lambda>`. Passing **behaviour** as data ("sort by this", "call this when done") removes families of near-duplicate functions.

How it works

```
yell = shout                                  # assign to a name
table = {"up": str.upper, "low": str.lower}  # store in a dict
print(apply_twice(shout, "hey"))              # pass as argument  -> HEY!!

nums = [5, -2, 8, -7, 3]
list(map(abs, nums))                  # [5, 2, 8, 7, 3]
list(filter(lambda n: n > 0, nums))   # [5, 8, 3]
[n * n for n in nums if n > 0]        # [25, 64, 9]
```

- `map(fn, it)` applies `fn` to each item; `filter(pred, it)` keeps items where the *predicate* (yes/no function) is truthy. Both return **lazy iterators** (values produced on request) — wrap in `list()`; consumable once.
- `sorted(it, key=fn, reverse=False)` returns a **new list**; `list.sort()` sorts in place, returns `None`. `min`, `max`, `heapq.nlargest`, `itertools.groupby` take `key=` too.
- `key` is called **once per element**; sorting is **stable** (equal keys keep order) — the basis of multi-level sorts (§5.7).
- `operator.itemgetter(1)`, `operator.attrgetter("age")`, `str.lower` are ready-made, faster alternatives to many lambdas.

```
people = [("Ana", 31), ("bob", 25), ("Cy", 31), ("dee", 19)]
sorted(people, key=lambda p: p[1])            # by age
sorted(people, key=lambda p: (-p[1], p[0]))   # age desc, then name
sorted(["bob", "Ana", "dee", "Cy"], key=str.lower)
max(people, key=operator.itemgetter(1))
```

```
by age:             [('dee', 19), ('bob', 25), ('Ana', 31), ('Cy', 31)]
by age desc, name:  [('Ana', 31), ('Cy', 31), ('bob', 25), ('dee', 19)]
case-insensitive:   ['Ana', 'bob', 'Cy', 'dee']
itemgetter(1):      ('dee', 19)
oldest:             ('Ana', 31)
```

A **tuple** key sorts by first element, then second…; negating a number flips that level.

Trade-offs

Tool	Best for	Watch out
Comprehension	Transform + filter in one line	Builds a whole list (generator `( … )` for laziness)
`map` / `filter` with a named function	`map(str.strip, lines)`	With a lambda, usually less readable than a comprehension
`lambda`	Tiny one-off `key=` functions	No statements; `<lambda>` in tracebacks; PEP 8: don't assign one to a name — use `def`

When not to use it

`f = lambda x: x + 1` → write `def` (better traceback name, docstring). Don't nest `map(lambda..., filter(lambda..., ...))` when a comprehension reads plainly.

Common mistakes

- `sorted_list = my_list.sort()` → `None`.
- Consuming a `map` / `filter` twice — second pass empty.
- `key=len()` (calling) instead of `key=len`.

Real-world examples

- pandas: `df["name"].map(str.title)`, `df.sort_values(key=...)`, `df.apply(fn)`.
- `sorted(files, key=os.path.getmtime)` (last-modified time).
- Callbacks: `threading.Thread(target=worker)`; `executor.map(fetch, urls)` in `concurrent.futures` (an *executor* is a worker pool running calls in parallel).

In the interview

> **30-second answer:** "Functions are objects, so they can be passed, stored and returned; a higher-order function takes or returns one. `lambda` makes a one-expression anonymous function, mostly for `key=`. `map` / `filter` return lazy iterators; comprehensions are usually preferred. `sorted` with `key` computes the key once per element and is stable, so tuple keys give multi-level sorts."

- *`sorted` vs `.sort()`?* → Any iterable → new list vs list method, in place, `None`.
- *Two fields, one descending?* → Tuple key with a negated number, or two stable sorts (secondary first).
- *`if` in a lambda?* → No statements; a conditional *expression* is fine.

TL;DR

Functions are values; higher-order functions take/return them. `lambda` for tiny keys. `map` / `filter` lazy; comprehensions clearer. `sorted(key=…)`: once per item, stable, tuple keys for multi-level.

7.6 Decorators and `functools`

What it is

A **decorator** takes a function and returns a replacement (usually a closure calling the original plus extra behaviour). `@` is shorthand:

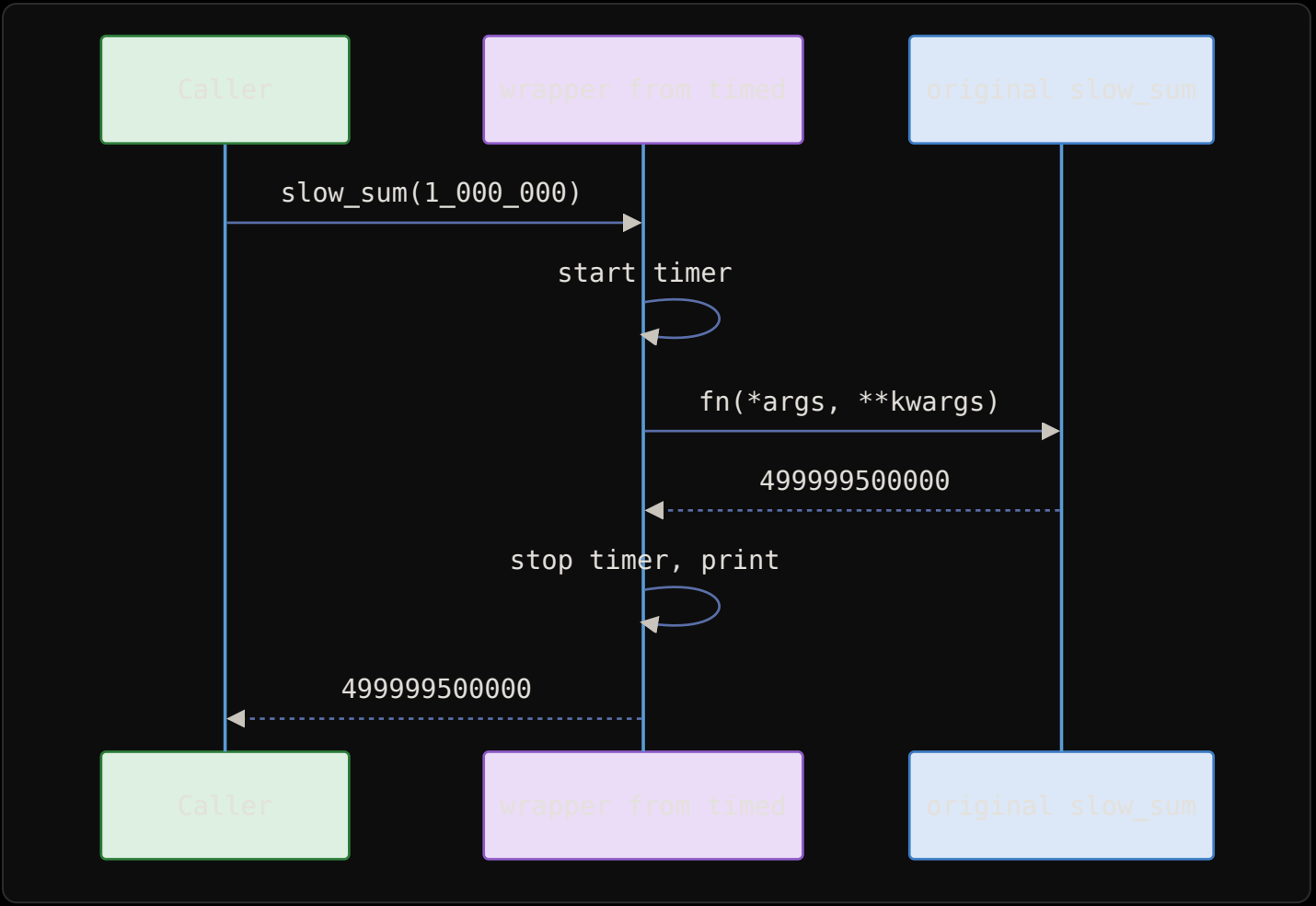
```
@timed
def slow_sum(n): ...
# is exactly
def slow_sum(n): ...
slow_sum = timed(slow_sum)
```

`functools` collects function tools: `wraps`, `partial`, `reduce`, `lru_cache`, `cache`, `cached_property`, `singledispatch`, `total_ordering`.

Why it matters

Decorators add cross-cutting behaviour (concerns spanning many functions) — timing, logging, caching, auth checks, retries, route registration — without editing each function; `functools` provides battle-tested versions.

How it works




```python
def timed(fn):
    @functools.wraps(fn)                # copy __name__, __doc__, etc.
    def wrapper(*args, **kwargs):       # accept any signature
        t0 = time.perf_counter()
        result = fn(*args, **kwargs)    # call the original
        ms = (time.perf_counter() - t0) * 1000
        print(f"  [{fn.__name__} ran in {'under' if ms < 500 else 'over'} 500 ms]")
        return result                   # don't forget to return!
    return wrapper
```

```
  [slow_sum ran in under 500 ms]
  slow_sum(1_000_000) = 499999500000
  with wraps:    name = slow_sum | doc = Add 0..n-1 the slow way.
  without wraps: name = wrapper | doc = None
```

`*args, **kwargs` let the wrapper accept and forward any signature.

`functools.wraps` copies `__name__`, `__doc__`, `__module__`, `__qualname__`, `__dict__` and sets `__wrapped__`; without it every decorated function appears as `wrapper` in logs, `help()` and tracebacks.

`functools.partial(fn, *args, **kwargs)` pre-fills arguments:

```
int_from_binary = functools.partial(int, base=2)
int_from_binary("1011")   # 11
```

`functools.reduce(fn, iterable[, initial])` folds pairwise left to right: `reduce(mul, [1,2,3,4,5])` = ((((1·2)·3)·4)·5) = 120. Without `initial`, empty input → `TypeError`. Prefer `sum`, `math.prod`, `max`, `"".join` where they fit.

`functools.lru_cache(maxsize=128)` memoizes: stores results keyed by arguments and returns them on repeat calls; beyond `maxsize` keys, the **least recently used** is evicted. **`functools.cache`** (3.9+) = `lru_cache(maxsize=None)` — unbounded, slightly faster.

```
@functools.lru_cache(maxsize=128)
def square_slowly(n):
    time.sleep(0.05)
    return n * n
for n in (4, 4, 5, 4):
    square_slowly(n)
print(square_slowly.cache_info())
```

```
CacheInfo(hits=2, misses=2, maxsize=128, currsize=2)
after cache_clear: CacheInfo(hits=0, misses=0, maxsize=128, currsize=0)
unhashable argument -> unhashable type: 'list'
```

4, 4, 5, 4 → misses on first 4 and 5, hits on the repeats. Arguments must be **hashable**.

Trade-offs

Tool	Win	Cost / caveat
`lru_cache(maxsize=N)`	Bounded memory	Bookkeeping; stale results if data changes
`cache`	Fastest, simplest	Grows forever — a memory leak for unbounded inputs
`partial`	Picklable, introspectable (`.func`, `.args`)	Only pre-fills; can't reorder
`reduce`	General fold	Often less readable than a loop or built-in

When not to use it

- Don't cache functions with side effects (sending email) or results depending on time, randomness or mutable globals.
- `lru_cache` on an **instance method** keys on `self` and keeps every instance alive; prefer `functools.cached_property` or a module-level cached function.

Common mistakes

- No `return result` in the wrapper → decorated function returns `None`.
- No `@functools.wraps`.
- `@timed()` vs `@timed` confusion. A decorator *with* arguments (`@retry(3)`) is a function returning a decorator — three levels of nesting.

Real-world examples

- Flask/FastAPI `@app.get("/items")` registers route handlers.
- pytest: `@pytest.fixture`, `@pytest.mark.parametrize(...)`.
- Standard library: `@property`, `@staticmethod`, `@classmethod`, `@dataclass`, `@contextlib.contextmanager`.
- `re` caches compiled patterns so repeated `re.match(pattern, s)` doesn't recompile.

In the interview

> **30-second answer:** "A decorator is a callable that takes a function and returns a replacement; `@d` above `def f` means `f = d(f)`. Inside you define a `wrapper(*args, **kwargs)` closure and use `functools.wraps` to keep the original's name and docstring. `functools` also gives `partial` to pre-fill arguments, `reduce` to fold, and `lru_cache` / `cache` to memoize pure functions (output depends only on inputs, no side effects) with hashable arguments."

- *`wraps`? →* Copies metadata (`__name__`, `__doc__`, …), sets `__wrapped__`.
- *`@retry(3)`? →* Outer function takes `3`, returns the decorator, which returns the wrapper.
- *When is `lru_cache` wrong? →* Impure functions, unhashable args, unbounded keys with `maxsize=None`, instance methods.

TL;DR

Decorator = function in, function out; always `@wraps` and `return result`. `partial` pre-fills, `reduce` folds, `lru_cache` (default `maxsize` 128) / `cache` memoize pure functions with hashable arguments.

7.7 Recursion

What it is

A **recursive function** calls itself on a smaller version of the problem — *divide and conquer*. It needs:

1. A **base case** — small enough to answer directly.
2. A **recursive case** — shrink the input and recurse, guaranteeing progress toward the base case.

Why it matters

Natural for **self-similar data**: nested lists/JSON, folder trees, parse trees, tree/graph traversal, merge sort and quicksort, backtracking (try an option, undo if it fails: permutations, sudoku). Interviews use it constantly.

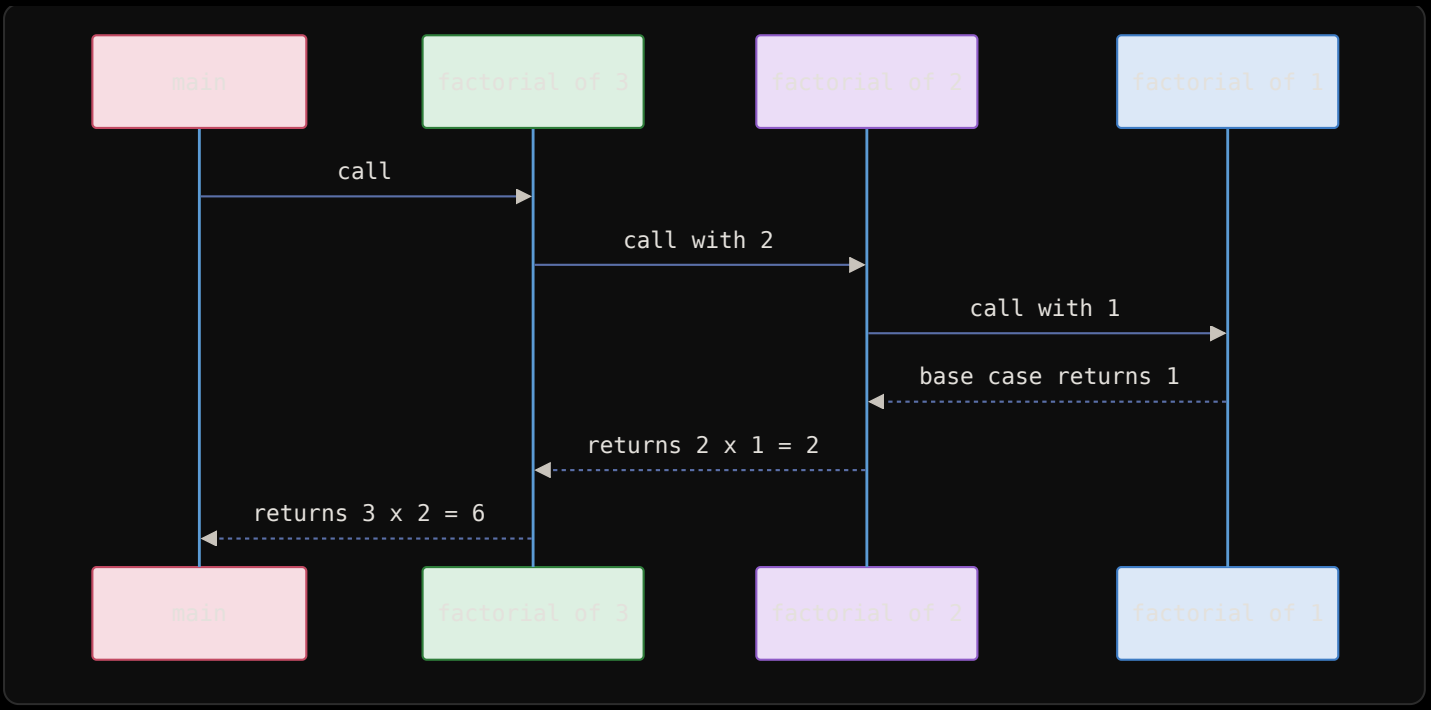
How it works — the call stack

Each call gets a frame on the **call stack** (last-in-first-out pile of active calls); frames pile up to the base case, then results flow back. Factorial: n! = n × (n−1)!, 0! = 1! = 1.

```python
def factorial(n):
    if n <= 1:                # base case (also handles 0! = 1)
        return 1
    return n * factorial(n - 1)
```

Trace of `factorial(3)`, and `factorial(5)` = 5 × 4 × 3 × 2 × 1 = 120:

```
factorial(5) = 120 | factorial(0) = 1
factorial(3)
  factorial(2)
    factorial(1)
    -> 1
  -> 2
-> 6
```



The recursion limit

CPython has **no tail-call optimisation** (TCO — reusing the frame when the last step is a self-call), by design to keep tracebacks complete, so every level costs a frame. Depth is capped: `sys.getrecursionlimit()` is **1000** by default; exceeding it raises `RecursionError` (a `RuntimeError` subclass).

```
sys.getrecursionlimit() = 1000
depth_of(500) = 500
RecursionError: maximum recursion depth exceeded
```

`sys.setrecursionlimit(n)` raises it, but deep recursion costs memory and can crash the process — for thousands of levels **rewrite as a loop** (explicit stack list for tree walks).

Memoization — fixing exponential recursion

Fibonacci: F(0)=0, F(1)=1, F(n)=F(n−1)+F(n−2); F(7) = 13. Naive recursion recomputes the same values (F(5) calls F(4) and F(3); F(4) calls F(3) *again*), so calls grow ~1.6^n. **Memoization** computes each F(k) once — linear work.

```python
@functools.cache
def fib_memo(n):
    if n < 2:
        return n
    return fib_memo(n - 1) + fib_memo(n - 2)

def fib_iter(n):              # bottom-up: no recursion at all
    a, b = 0, 1
    for _ in range(n):
        a, b = b, a + b
    return a
```

```
fib(7) = 13
naive calls for n=10, 20, 25: 177 21891 242785
memoized: CacheInfo(hits=23, misses=26, maxsize=None, currsize=26)
fib_iter(90) = 2880067194370816120
```

n = 25: 242,785 naive calls vs 26 distinct values computed (misses) and 23 cache hits.

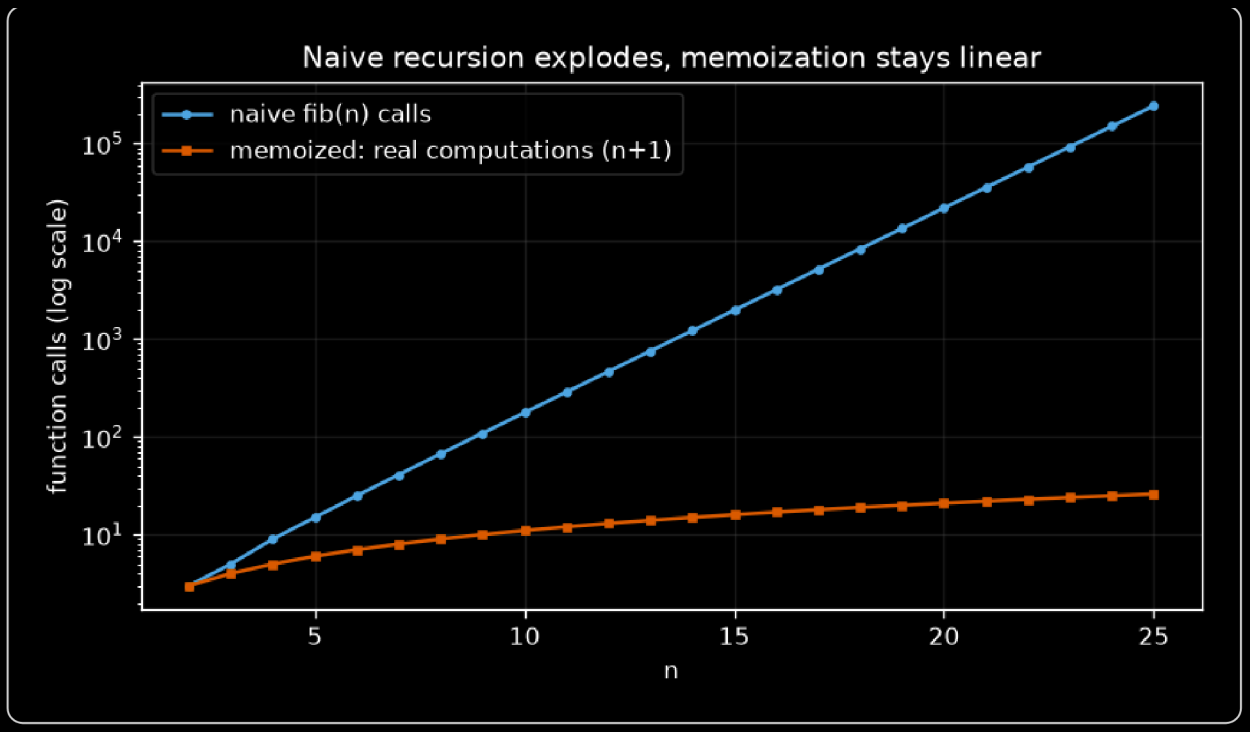


Figure — calls for fib(n): naive grows exponentially (a straight line on a log axis), memoized linearly.

Recursion over nested data

```python
def flatten(nested):
    out = []
    for item in nested:
        if isinstance(item, list):
            out.extend(flatten(item))   # recursive case: a smaller list
        else:
            out.append(item)            # base case: a plain value
    return out

flatten([1, [2, [3, [4]], 5], [], 6])   # [1, 2, 3, 4, 5, 6]
```

The same shape walks directories (`os.walk` / `Path.iterdir`), JSON and trees.

Trade-offs

	Recursion	Iteration
Fits	Trees, nested data, divide and conquer, backtracking	Linear sequences, counting, very deep processes
Clarity	Mirrors the definition	May need an explicit stack
Memory	One frame per level	Constant (or your own stack)
Depth limit	~1000 by default	None
Speed in CPython	Call overhead per level	Usually faster

When not to use it

Linear problems (sum a list, walk a 100k-node linked list) — loop. Depth driven by untrusted input size.

Common mistakes

- **Missing/unreachable base case**: base `n == 1` makes `factorial(0)` recurse 0 → −1 → … until `RecursionError`; `n <= 1` fixes it.

- Not shrinking the input (`f(n)` calling `f(n)`).
- Forgetting to `return` the recursive result → `None`.
- Naive recursion on overlapping sub-problems (the same smaller problems solved repeatedly: Fibonacci, grid paths) — memoize.
- Mutable default accumulators (`def walk(node, seen=set())`).

Real-world examples

- `os.walk`, `shutil.copytree`, `json` encoding (CPython's encoder is recursive and can raise `RecursionError` on extremely deep input), `copy.deepcopy`.
- Recursive-descent parsers (one function per grammar rule, calling each other).

In the interview

> **30-second answer:** "Recursion is a function calling itself on a smaller input until a base case. Each call uses a stack frame; CPython doesn't optimise tail calls and limits depth to about 1000, raising `RecursionError`. Overlapping sub-problems like Fibonacci are exponential unless memoized with `functools.cache`, which makes them linear. For very deep or linear problems, convert to a loop."

- *Naive vs memoized Fibonacci?* → O(φⁿ) (φ ≈ 1.618, golden ratio) time vs O(n) time and O(n) space.
- *Tail-call optimisation?* → No, by design.
- *Recursive DFS (depth-first search) to iterative?* → Explicit stack: pop a node, process, push children.

TL;DR

Base case + shrinking recursive case. Frames pile up; limit ≈ 1000; no TCO. Memoize overlapping sub-problems (`@cache`); loop when deep.

7.8 Docstrings and annotations

What it is

- A **docstring** is a string literal as the **first statement** of a function, class or module, stored in `__doc__` and shown by `help()`, IDEs and documentation generators (Sphinx). PEP 257: one-line summary, blank line, details.
- **Annotations** (type hints) declare expected types: `def f(x: int) -> str`. **Not enforced** at runtime — stored and ignored. **mypy**, **pyright** (static type checkers) and IDEs read them to catch bugs before running; **FastAPI** (web framework) and **Pydantic** (data validation) read them at runtime to validate. Full coverage: Chapter 13.

Why it matters

Docstrings explain *what and why*; annotations make the contract checkable, so a type checker catches `average("abc")` in CI (automated checks on every change).

How it works

```python
from collections.abc import Callable, Iterable

def average(values: Iterable[float], *, default: float | None = None) -> float | None:
    """Return the arithmetic mean of *values*.

    Args:
        values: numbers to average.
        default: returned when *values* is empty.

    Returns:
        The mean, or *default* for empty input.

    >>> average([1, 2, 3])
    2.0
    """
    items = list(values)
    if not items:
        return default
    return sum(items) / len(items)

def apply(fn: Callable[[int], int], x: int) -> int:
    return fn(x)

type Pair = tuple[int, int]          # 3.12+ type alias statement
def swap(p: Pair) -> Pair:
    return p[1], p[0]
```

```
first docstring line: Return the arithmetic mean of *values*.
signature: (values: collections.abc.Iterable[float], *, default: float | None = None) -> float | None
return annotation: float | None
average([1, 2, 3]) = 2.0
average([], default=0.0) = 0.0
swap(('x', 'y')) = ('y', 'x') <- str, not int, and no error
apply(lambda n: n + 1, 41) = 42
swap((1, 2)) = (2, 1)
doctest: 1 test(s), 0 failure(s)
```

Syntax	Meaning	Since
`x: int`, `-> str`	Parameter / return annotation	3.0 (PEP 3107); typing semantics 3.5 (PEP 484)
`list[int]`, `dict[str, float]`	Built-in generics (containers with a type parameter)	3.9
`int | None`	Union ("either type"; replaces `Optional[int]`)	3.10
`type Alias = ...`, `def f[T](x: T) -> T`	Type alias / generic syntax	3.12 (PEP 695)
Lazy evaluation of annotations	Forward references (types defined later) work without quotes	3.14 (PEP 649)

`inspect.signature(fn)` reads parameters, defaults and annotations at runtime. The `>>>` lines form a **doctest**: `doctest.testmod()` runs them and compares output.

Trade-offs

Annotations add noise and need a checker to pay off, but on shared code they catch whole bug classes and power autocompletion. Docstrings drift — keep them short and about behaviour.

When not to use it

Throwaway scripts needn't be fully annotated; don't restate the obvious (`"""Returns x."""`).

Common mistakes

- Believing annotations validate input — `swap(("x", "y"))` ran fine.
- A `# does X` comment instead of a docstring → invisible to `help()`.
- Docstring not *first* → `__doc__` is `None`.

Real-world examples

- FastAPI builds request validation and OpenAPI docs (machine-readable API description) from annotations.
- NumPy/Google docstring styles rendered by Sphinx (numpy.org).
- Dropbox runs mypy in CI over millions of lines.

In the interview

> **30-second answer:** "A docstring is the first string in a function, stored in `__doc__` and shown by `help()`. Annotations declare expected types but aren't enforced at runtime; static checkers like mypy use them, and frameworks like FastAPI/Pydantic read them to validate."

- *Checked at run time?* → No, only by tools/libraries that read them.
- *Read annotations at runtime?* → `inspect.signature(f)`, `inspect.get_annotations(f)` / `f.__annotations__`.

TL;DR

Docstring = first-statement string shown by `help()`. Annotations = documentation checkable by mypy/pyright, not enforced at runtime.

Chapter TL;DR

- `def` creates a function object; calling runs it in a fresh frame; no `return` ⇒ `None`; `pass` for empty bodies.
- Arguments: positional (order) and keyword (name); defaults evaluated **once** ⇒ never a mutable default, use `None`; `*args` tuple, `**kwargs` dict; `/` positional-only, `*` keyword-only; `*`/`**` unpack at call sites; `return a, b` returns a tuple.
- Scope: **LEGB**. Assignment makes a name local for the whole function (`UnboundLocalError`). `global` → module, `nonlocal` → enclosing function. `if`/`for` don't create scopes.
- Closures remember enclosing variables (by reference ⇒ late binding). Decorators are closures: `@d` means `f = d(f)`; use `functools.wraps`.
- Functions are first-class: pass them, return them, `key=` them. `map`/`filter` are lazy; `sorted` is stable.
- `functools`: `partial`, `reduce`, `lru_cache`/`cache` (pure functions, hashable args), `wraps`.
- Recursion: base case + progress; limit 1000, no tail-call optimisation; memoize overlapping sub-problems; loop when deep.
- Docstrings for `help()`, annotations for tools — not runtime checks.